

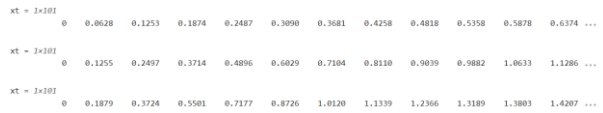


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Program:	BSCpE	Section:	A461

A. Guided Exercises:

(Please include the following: 1. Photos of the codes and 2. Outputs after executing the MATLAB code.)

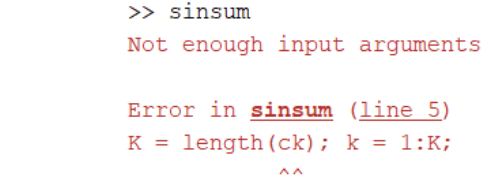
Exercise 1

Photo/s of MATLAB output	Photos of MATLAB Codes
 <pre> xt = 1x101 0 0.0628 0.1253 0.1874 0.2487 0.3090 0.3681 0.4258 0.4818 0.5358 0.5878 0.6374 ... xt = 1x101 0 0.1255 0.2497 0.3714 0.4896 0.6029 0.7104 0.8110 0.9039 0.9882 1.0633 1.1286 ... xt = 1x101 0 0.1879 0.3724 0.5501 0.7177 0.8726 1.0120 1.1339 1.2366 1.3189 1.3803 1.4207 ... </pre>	Exercise 1: Sum of Sinusoidal Functions. Generate the sample of $x(t)$ at time instances 0:0.01:1. $x(t) = \sin(2\pi t) + \frac{1}{3} \sin(6\pi t) + \frac{1}{5} \sin(10\pi t) = \sum_{k=1}^3 \frac{1}{k} \sin(2\pi k t), \quad 0 \leq t \leq 1$ <pre> t=0:0.01:1; xt=zeros(1,length(t)); for k=1:3 xt = xt + (1/k)*sin(2*pi*k*t) end </pre>

Exercise 2

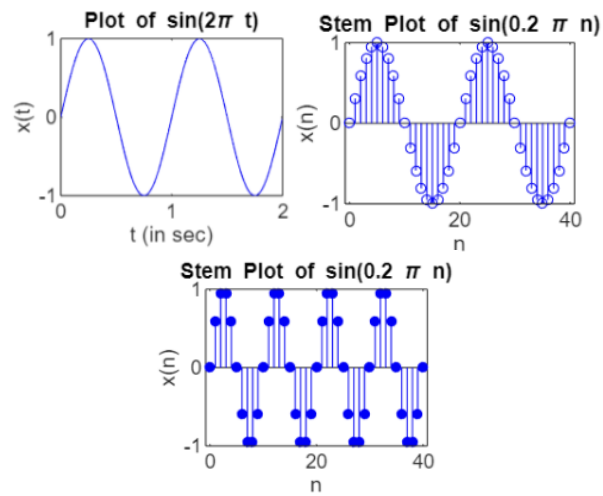
Photo/s of MATLAB output	Photos of MATLAB Codes
 <pre> xt = 1x101 0 0.0628 0.1253 0.1874 0.2487 0.3090 0.3681 0.4258 0.4818 0.5358 0.5878 ... xt = 1x101 0 0.1255 0.2497 0.3714 0.4896 0.6029 0.7104 0.8110 0.9039 0.9882 1.0633 ... xt = 1x101 0 0.1879 0.3724 0.5501 0.7177 0.8726 1.0120 1.1339 1.2366 1.3189 1.3803 ... </pre>	Exercise 2: Sum of Sinusoidal Functions. Generate the sample of $x(t)$ at time instances 0:0.01:1. $x(t) = \sum_{k=1}^K c_k \sin(2\pi k t)$ <pre> t=0:0.01:1; k=1:2:5; ck = 1./k; xt = ck * sin(2*pi*k'*t); </pre>

Exercise 3

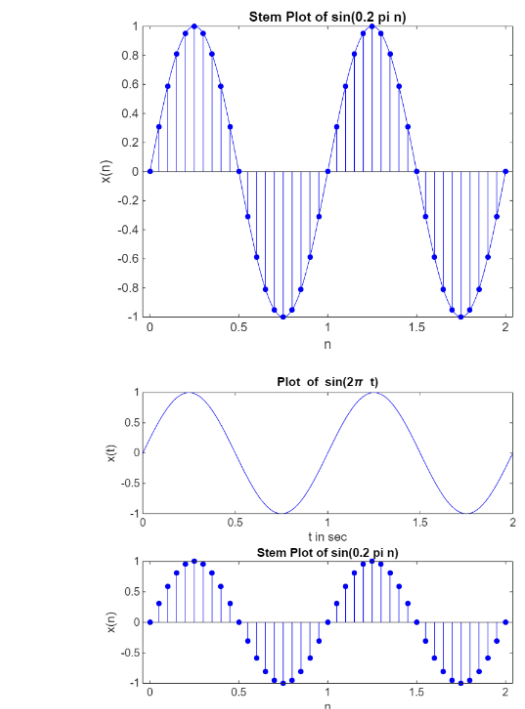
Photo/s of MATLAB output	Photos of MATLAB Codes
 <pre> >> sinsum Not enough input arguments. Error in sinsum (line 5) K = length(ck); k = 1:K; ^^ </pre>	<pre> function xt = sinsum(t,ck) % Computes sum of sinusoidal terms of the form in (1.1) % x = sinsum(t,ck) % K = length(ck); k = 1:K; ck = ck(:)'; t = t(:)'; xt = ck * sin(2*pi*k'*t); end </pre>



Exercise 4

Photo/s of MATLAB output	Photos of MATLAB Codes
	<pre> t = 0:0.01:2; % sample points from 0 to 2 in steps of 0.01 x = sin(2*pi*t); % Evaluate sin(2 pi t) plot(t,x,'b'); % Create plot with blue line xlabel('t (in sec)'); %Label axis ylabel('x(t)'); % Label axis title('Plot of sin(2\pi t)'); % Title plot n = 0:1:40; % sample index from 0 to 20 x = sin(0.1*pi*n); % Evaluate sin(0.2 pi n) Hs = stem(n, x, 'b'); % Stem-plot with handle Hs set(Hs, 'markersize', 4); % Change circle size xlabel('n'); % Label axis ylabel('x(n)'); % Label axis title('Stem Plot of sin(0.2 \pi n)'); % Title plot n = 0:1:40; % sample index from 0 to 20 x = sin(0.2*pi*n); % Evaluate sin(0.2 pi n) Hs = stem(n, x, 'b', 'filled'); % Stem-plot with handle Hs set(Hs, 'markersize', 4); % Change circle size xlabel('n'); ylabel('x(n)'); % Label axis title('Stem Plot of sin(0.2 \pi n)'); %Title plot </pre>

Exercise 5

Photo/s of MATLAB output	Photos of MATLAB Codes
	<pre> t=0:0.01:2; % sample points from 0 to 2 in steps of 0.01 x=sin(2*pi*t); % Evaluates sin(2 pi t) plot(t,x,'b'); % Create plot with blue line hold on; xlabel('t in sec'); ylabel('x(t)'); % Label axis title('Plot of sin(2\pi t)'); % Title plot n = 0:1:40; % sample index from 0 to 20 x = sin(0.1*pi*n); % Evaluate sin(0.2 pi n) Hs = stem(n*0.05,x,'b','filled'); % Stem-plot with handle Hs set(Hs, 'markersize',4); % Change circle size hold off; xlabel('n'); ylabel('x(n)'); % Label axis title('Stem Plot of sin(0.2 \pi n)'); % Title plot t=0:0.01:2; % sample points from 0 to 2 in steps of 0.01 x=sin(2*pi*t); % Evaluates sin(2 pi t) subplot(2,1,1); % two rows, one column, first plot plot(t,x,'b'); % Create plot with blue line hold on; xlabel('t in sec'); ylabel('x(t)'); % Label axis title('Plot of sin(2\pi t)'); % Title plot n = 0:1:40; % sample index from 0 to 20 x = sin(0.1*pi*n); % Evaluate sin(0.2 pi n) subplot(2,1,2); %two rows, one column, second plot Hs = stem(n*0.05,x,'b','filled'); % Stem-plot with handle Hs set(Hs, 'markersize',4); % Change circle size hold off; xlabel('n'); ylabel('x(n)'); % Label axis title('Stem Plot of sin(0.2 \pi n)'); % Title plot </pre>

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B. Problem Exercises:

(Please include the following: 1. Photos of the codes and 2. Outputs after executing the MATLAB code.)

Problem 1:

Photo/s of MATLAB output	Photos of MATLAB Codes
(Insert image/s here) a) <pre>z = 1x11 0 0.0199 0.0779 0.1694 0.2869 0.4207 0.5592 ...</pre> b) <pre>k = 1x11 0 0.0001 0.0016 0.0081 0.0256 0.0625 0.1296 ...</pre> c) <pre>l = 4.8093</pre>	<pre>% Problem 1 x = 0:0.1:1; % Vector x y = sin(0:0.2:2); % Vector y % a) Element-wise multiplication z = x .* y % b) Element-wise multiplication of x four times k = x .* x .* x .* x % c) Dot product of x and y l = dot(x, y)</pre>

Problem 2:

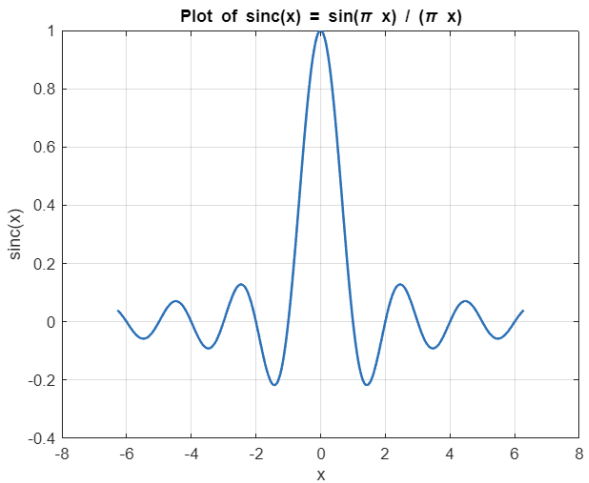
Photo/s of MATLAB output	Photos of MATLAB Codes
a) <pre>size_A = 1x2 3 3</pre> b) <pre>size_B = 1x2 2 3</pre> c) <pre>C = 3x2 4 1 8 5 7 4</pre> d) <pre>D = 3x3 12 2 -6 3 8 4 -6 12 18</pre> e) <pre>det_A = 492</pre> f) <pre>A_squared = 3x3 186 36 -144 -32 118 300 -172 86 408</pre> g) <pre>AB = 3x2 30 3 156 90 134 86</pre>	<pre>A = [12 3 -6; 2 8 12; -6 4 18]; B = [4 8 7; 1 5 4]; % a) Size of each matrix size_A = size(A) size_B = size(B) % b) Transpose of B into C C = B.' % c) Transpose of A into D D = A.' % d) Determinant of A det_A = det(A) % e) Determinant of C det_C = det(C) % f) A squared A_squared = A^2 % g) Multiply A and B AB = A * B</pre>

INTRODUCTION TO MATLAB

Problem 3:

Photo/s of MATLAB output	Photos of MATLAB Codes
a) $z1 = 45$ $z2 = 720$ $z3 = 135$	a) <pre>% a) Verify the script z1 = degTorad(pi/4) z2 = degTorad(4*pi) z3 = degTorad(3*pi/4)</pre>
b) $rad1 = 45$ $rad2 = 720$ $rad3 = 135$	b) <pre>% b) Use MATLAB built-in rad2deg function to verify rad1 = rad2deg(pi/4) rad2 = rad2deg(4*pi) rad3 = rad2deg(3*pi/4)</pre>

Problem 4:

Photo/s of MATLAB output	Photos of MATLAB Codes
 Plot of $\text{sinc}(x) = \sin(\pi x) / (\pi x)$	<pre>function plot_sinc() % Define x range x = linspace(-2*pi, 2*pi, 1000); % sinc(x) = sin(pi*x)/(pi*x) y = sin(pi*x) ./ (pi*x); % Handle the case when x = 0 y(x == 0) = 1; % Plot plot(x, y, 'LineWidth', 1.5); grid on; xlabel('x'); ylabel('sinc(x)'); title('Plot of sinc(x) = sin(\pi x) / (\pi x)'); end</pre> <pre>% Problem 4 plot_sinc();</pre>

C. Findings ,Observations, and Comments:*Guiding Questions :*

1. *What did you learn in the guided experiments? Did you learn something new?*
2. *What are your findings, observations, and comments in the guided exercises?*
3. *Did you solve the problem exercises?*
4. *Are there any difficulties during coding or set-up that you encountered?*
5. *What are your findings, observations, and comments in the problem exercise?*

In the guided experiments, I learned how to use basic MATLAB commands and functions to perform calculations, create variables, and display outputs. I also became more familiar with the MATLAB interface and how to write simple scripts. This was new to me because I had little experience with MATLAB before, and now I understand how it works for basic programming and computation tasks.

I observed that MATLAB is very useful for performing mathematical operations quickly and accurately. It can also display outputs in different formats, which makes it easier to interpret results. I noticed that once I understood the commands, the process became faster and more efficient. The exercises were clear and helpful for practicing the basics.

Yes, I was able to solve the problem exercises. Some problems required more time because I needed to figure out the correct syntax and commands, but with careful checking, I was able to get the correct outputs.

At first, I had difficulty remembering the correct syntax for some MATLAB commands, which caused errors when I ran the code. I also had to make sure I was typing the commands exactly as needed because even small mistakes could stop the program from running. However, after rechecking and testing step-by-step, I was able to fix the issues.

In the problem exercises, I found that applying what I learned from the guided activities made the solutions easier to figure out. I also observed that MATLAB is very powerful for solving numerical problems quickly. The exercises helped me understand how to break down a problem into smaller steps and use MATLAB functions to solve each part. Overall, the activities improved my problem-solving skills and made me more confident in using MATLAB.